Hall Ticket Number:

MR-21

Code: B313CS5

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular Examinations, March-2023

Discrete Mathematics

(Common to CSE, CSM & CSD)

Time: 3 Hours

Answer ALL Questions

Max. Marks: 70

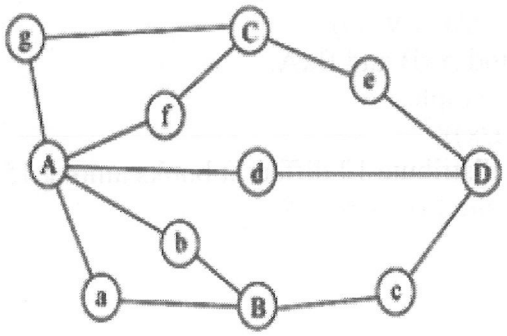
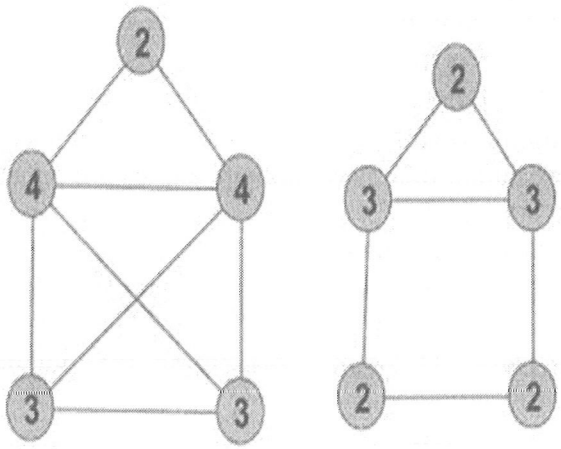
Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define a proposition and give an example?	2	1	1	1
1. b)	Construct the truth table for ~PΛ(~PV~Q).	2	2	1	1,2
1. c)	A={a, b, c}, B={1, 2} then find AxB and BxA.	2	2	2	1
1. d)	What is a digraph? Give an example.	2	1	2	1
1. e)	Find the middle term of $(x+(1/x))^{20}$.	2	2	3	1
1. f)	How many ways are there to distribute 12 different books among 15. people if no person receive more than 1 book?	2	2	3	1
1. g)	Solve $\mathbf{a_n-6a_{n-1}+9a_{n-2}=0}$.	2	3	4	1,2
1. h)	Find the generating function of 1, 2, 3, 4…..	2	2	4	1
1. i)	Define a planar graph?	2	1	5	1
1. j)	List the applications of Trees?	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Obtain the PDNF of (PΛQ)V(~PΛR)V(QΛR) without using truth table.	5	3	1	1,2
2. b)	Show that (PΛQ) → (PVQ) is a tautology.	5	3	1	1,2
	OR				
2. c)	Show that (~PΛ(~QΛR))V(QΛR)V(PΛR) ⇔R.	5	3	1	1,2
2. d)	Determine the validity of the following argument: If two sides of a triangle are equal, then two opposite angles are equal. Two sides of a triangle are not equal. Therefore, the opposite angles are not equal.	5	4	1	1,2
3. a)	Prove that a relation R from A to B and S be a relation from B to C, then $\mathbf{(RoS)^{-1}=S^{-1}oR^{-1}}$	5	3	2	1
3. b)	What is a Poset? Draw Hasse Diagram representing the positive divisors of 36.	5	2	2	1
	OR				
3. c)	A function f(x) = 4x+2 on set of real numbers is bijective or not. Also find inverse of it.	5	2	2	1
3. d)	Show that the relation R is defined NxN by (a, b)R(c, d) if and only if **a+d = b+c** is an equivalence relation.	5	3	2	1
4. a)	How many numbers of 4 digits can be formed from 0, 1, 2, 3, 5, 6 if no digit is to occur more than once in each number? How many of these are divisible by 2?	5	2	3	1
4. b)	Prove that $\mathbf{1^2+2^2+3^2+...+n^2= n(n+1)(2n+1)/6}$ using mathematical induction where n is a positive integer.	5	4	3	1
	OR				
4. c)	Find the coefficient of x^{10} in $\mathbf{(1+x+x^2+....)^2}$	5	2	3	1
4. d)	How many possible telephone numbers are there when there are seven digits, the first two of which are between 2 and 9 inclusive, the third digit between 1 and 9 inclusive, and each of the remaining may be between 0 and 9 inclusive?	5	2	3	1

P.T.O.

5. a)	In a neighbourhood, 90% children were falling sick due flu and 10% due to measles and no other disease. The probability of observing rashes for measles is 0.95 and for flu is 0.08. If a child develops rashes, find the child's probability of having flu.	5	3	4	1,2
5. b)	Solve recurrence relations by Generating Functions $a_n - 9a_{n-1} + 20a_{n-2} = 0$ for $n \geq 2$ and $a_0 = -3$, $a_1 = -10$	5	3	4	1,2
	OR				
5. c)	Solve the recurrence relations using the characteristic roots $a_n - 4a_{n-1} - 12a_{n-2} = 0$ for $n \geq 2$ and $a_0 = 4$, $a_1 = 16/3$	6	3	4	1,2
5. d)	What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7?	4	2	4	1
6. a)	Find the spanning tree using BFS algorithm 	6	2	5	1
6. b)	Define the following with an example: i) Chromatic Number ii) Euler Path iii) Complete graph Kn iv) Bipartite Graph	4	1	5	1
	OR				
6. c)	Show that the following graphs are Hamiltonian graphs. 	5	3	5	1,2
6. d)	State and prove the first theorem of graph theory.	5	2	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome